

Reasoning Under Social Uncertainty: Evaluating NARS in Multi-Agent
Information Environments

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Introduction

A core challenge in artificial general intelligence (AGI) is not only reasoning about the world, but reasoning about information sources that may be incomplete, misleading, or strategically deceptive. In many real-world settings, an intelligent agent does not observe facts directly, but instead, it must rely on statements from other agents, each of whom may vary in accuracy, honesty, and consistency. The problem addressed in this project is how an AI system can use socially sourced information to form beliefs while continuously updating its confidence in both the content of the information and the reliability of the source. This becomes especially important in environments where some agents are truthful, some are consistently deceptive, and others alternate between correct and incorrect claims. Under these conditions, the agent must infer not only what is true about the environment, but also who can be trusted and when trust should be even revised. The project therefore focuses on the robustness of reasoning under uncertainty in a socially mediated, and adversarial setting. More broadly, this would connect to central questions in AGI research about how a general intelligence should integrate evidence, manage uncertainty, and avoid being misled by unreliable testimony. A system that can reason effectively in this kind of environment would demonstrate a significant step toward the flexible, context-sensitive judgment expected of more general intelligence. For these reasons, the scenario serves as a useful test case for examining the strengths and limitations of particularly non-axiomatic reasoning in socially complex environments with the use of NARS (Non-Axiomatic Reasoning System).

The motivation for this project emerged naturally while learning NARS and building a mini-NARS implementation in one of the previous assignments for the course. As I worked through the system's core components, it became clear that one of the most interesting features of NARS is its emphasis on reasoning under insufficient and uncertain knowledge rather than assuming perfect access to truth. That perspective led to an important question: how would NARS behave when the uncertainty does not come only from the environment itself, but from other agents deliberately or inconsistently shaping the information it receives? In practice, social reasoning is a major part of intelligence, since humans constantly evaluate testimony, compare conflicting reports, and revise trust based on experience. This made the idea of a social manipulation scenario especially compelling to me, because it extends NARS from isolated inference tasks into a setting where source credibility becomes an essential part of the reasoning process. Another reason for pursuing this topic is that deceptive information environments are increasingly relevant in modern AI discussions, including misinformation, adversarial inputs, and uncertain multi-agent interaction. Although these conversations are typically held with Artificial Narrow Intelligence (ANI), studying how NARS responds to these conditions helps connect the theory of non-axiomatic reasoning to practical concerns about robustness and adaptability. Ultimately, this project was motivated by both intellectual curiosity and the broader goal of understanding whether a general reasoning system can learn whom to trust without relying on fixed assumptions or even perfect data.

Experiment Design

The central goal of this project is to test whether NARS can correctly infer objective environmental facts while simultaneously learning to distinguish trustworthy information sources from deceptive or inconsistent ones. To accomplish this, the experiment will use a simulated multi-agent environment containing a fixed set of objective ground-truth propositions. These propositions represent simple binary facts about the environment, such as whether an object exists in a particular location or whether a certain condition holds. Each fact remains constant throughout an experimental run to provide a stable reference against which NARS's evolving beliefs can be measured. This design choice isolates the challenge of social uncertainty from environmental instability, ensuring that any reasoning errors can be attributed primarily to source reliability rather than changes in the underlying world state. By keeping the environment itself static, the experiment focuses directly on how effectively NARS processes conflicting testimony. The reasoning challenge therefore becomes epistemic rather than perceptual: the system must determine what is true despite receiving potentially unreliable evidence. This provides a controlled framework for evaluating trust adaptation and belief revision under uncertain informational conditions.

To introduce varying levels of informational reliability, the environment includes four agent types that generate statements about the underlying facts. Truthful agents always provide statements that align with the ground truth, serving as reliable sources of evidence. Deceptive agents always provide false statements, intentionally contradicting reality to test NARS's resistance to misinformation. Noisy agents produce statements randomly, creating uncertainty without deliberate adversarial intent. Strategic agents present the most complex challenge by alternating between truthful and deceptive behavior according to either predefined rules or probabilistic patterns. This diversity of agent behaviors is intended to simulate a socially complex information environment in which source reliability cannot be assumed and must instead be inferred through repeated interaction. A reliability score is a numerical summary of how dependable an information source is. In this experiment, it represents the proportion of a source's claims that match the ground truth, or more generally how strongly the source can be expected to provide correct information. It also should be noted that NARS will not be provided with explicit reliability scores for any agent. Instead, the system must infer trustworthiness solely through observed consistency between incoming claims and previously accumulated evidence. So this all means that implicit reliability scores are associated with each agent type in order to establish clear expectations for source behavior. Truthful agents correspond to near-perfect (1.0) reliability meaning all their statements match ground truth, deceptive agents to near-zero reliability (0.0), and noisy agents to approximately neutral reliability (0.5) due to their random outputs. Strategic agents occupy an intermediate and fluctuating reliability space that changes over time, making them particularly difficult to classify. These distinctions are intentionally hidden from NARS so that any successful trust differentiation emerges from the reasoning process itself rather than from any external guidance. Statements delivered to the system contain three key components: the proposition itself, the truth value as claimed by the agent, and the identity of the source. This structure ensures that NARS receives sufficient contextual information to reason both about environmental facts and about source credibility. The sequential delivery and presentation of statements allows observation of how beliefs evolve incrementally rather than only examining final outcomes. This mirrors real-world reasoning scenarios in which information arrives over time and trust must be continuously updated. By structuring the experiment as a temporal stream of evidence, the design captures the dynamic nature of social reasoning under uncertainty.

System performance will be evaluated using four primary metrics: (1) belief accuracy, (2) confidence evolution, (3) trust inference, and (4) misinformation resistance. Belief accuracy measures how often NARS's inferred conclusions match the actual ground-truth facts, providing a direct indicator of reasoning effectiveness. Tracking this metric over time makes it possible to assess the speed and stability with which the system converges toward correct conclusions. Confidence evolution refers to examining how NARS adjusts its confidence values as additional evidence is received, revealing whether the system appropriately strengthens justified beliefs and weakens unsupported ones. This metric is particularly important because non-axiomatic reasoning depends not only on producing conclusions, but on assigning confidence proportional to available evidence. Trust inference evaluates whether NARS successfully differentiates between reliable and unreliable agents by examining how strongly it incorporates information from each source as interactions continue. Misinformation resistance specifically measures the system's ability to preserve accurate beliefs even when exposed to sustained false or misleading inputs from deceptive and strategic agents. A high score on this metric would indicate that NARS is capable of discounting harmful information rather than allowing incorrect testimony to dominate its reasoning process.

The experiment was implemented with python as a timed multi-agent simulation that generates a stream of statements from a fixed synthetic environment and then feeds those statements into NARS one at a time. In the main NARS run, the system creates 10 environment facts and a pool of 12 agents: three truthful, three deceptive, three noisy, and three strategic, and uses a fixed random seed so the run is reproducible. The simulation is then executed for 24 steps, and at each step every agent produces one statement, so the full run contains 288 total judgments before any are sent to NARS. Each statement records the simulation step, source agent, proposition, claimed truth value, and actual truth value, and it is converted into a Narsese judgment with a confidence value of 0.90 before being transmitted. In the OpenNARS-for-Applications (the zip I downloaded for using OpenNARS) version of the pipeline, these judgments are sent through a subprocess bridge using the NAR shellcommand, and they are fed step-by-step so the system can observe how belief confidence changes after each individual input rather than only at the end of the run. After each statement is sent, the code reads NARS's output, logs the latest belief state, and records whether the current judgment appears to match the environment, which makes it possible to analyze confidence evolution and source-learning dynamically over time.

I hypothesize that NARS will maintain relatively high belief accuracy even when it is exposed to misleading, inconsistent, and strategically deceptive information. I also expect it to gradually distinguish reliable sources from unreliable ones, assigning greater influence to agents whose claims repeatedly align with the ground truth. My reasoning for this is based on the way NARS is designed to reason under uncertainty rather than assume perfect or complete information. Because NARS updates beliefs incrementally and revises confidence as new evidence arrives, it should be able to reduce the impact of false testimony over time. In other words, repeated consistency should strengthen trust, while repeated contradiction should weaken it. I formulated this hypothesis because the project is not only testing whether NARS can identify correct facts, but also whether it can adapt to the social context in which those facts are communicated. In a deceptive multi-agent setting, robust reasoning depends on both factual inference and source evaluation. If NARS performs well, that would suggest that non-axiomatic reasoning can support resilient decision-making in environments where information cannot be assumed to be trustworthy, which makes the experiment valuable for observing whether NARS can handle a more realistic form of uncertainty than isolated, source-free inference.

Experiment Results

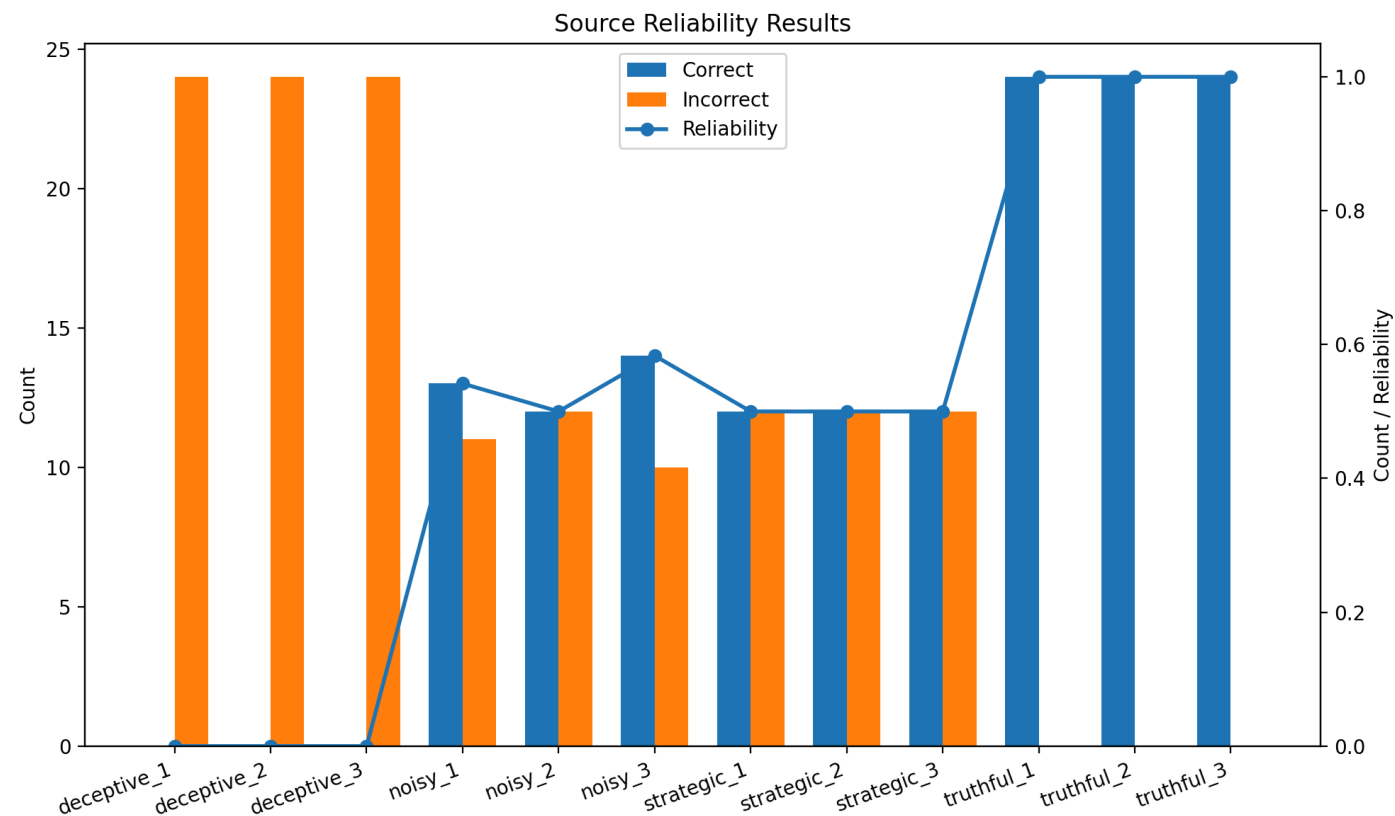


Figure 1.0

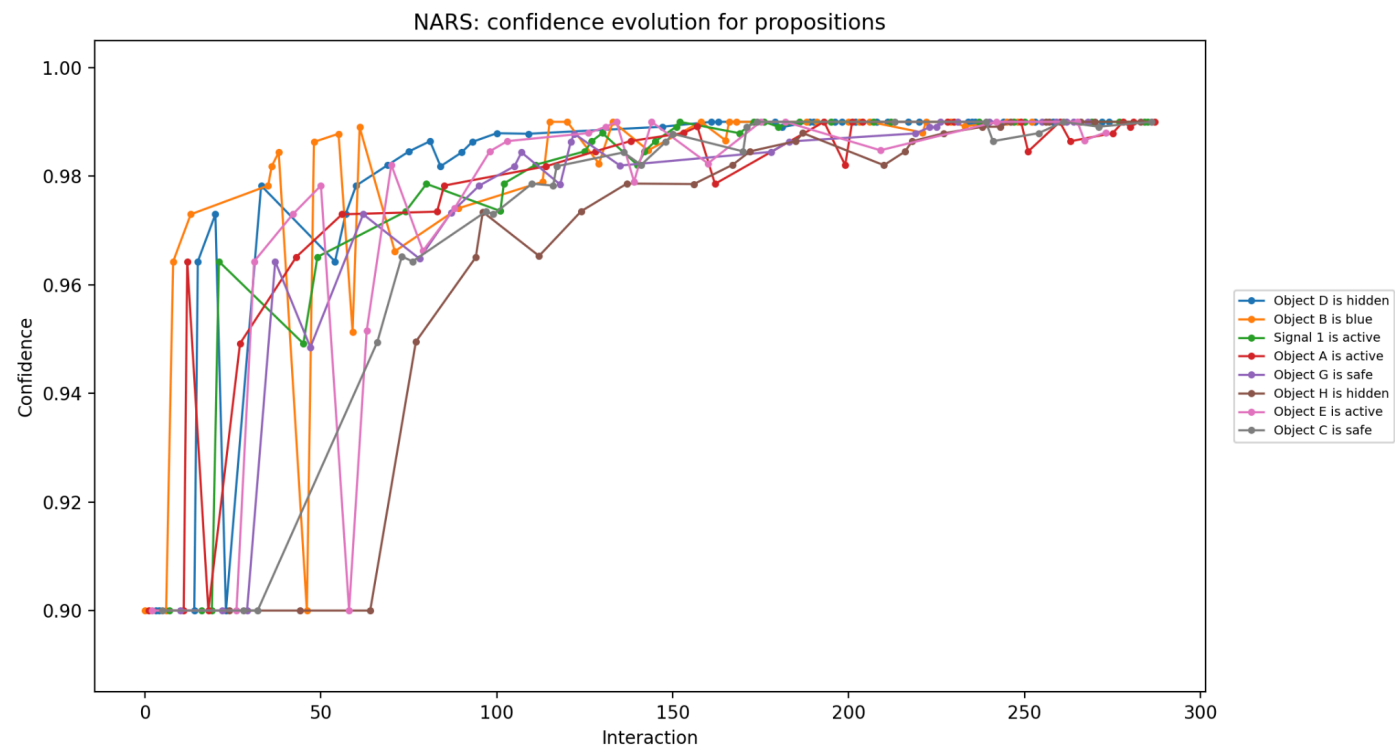


Figure 1.1

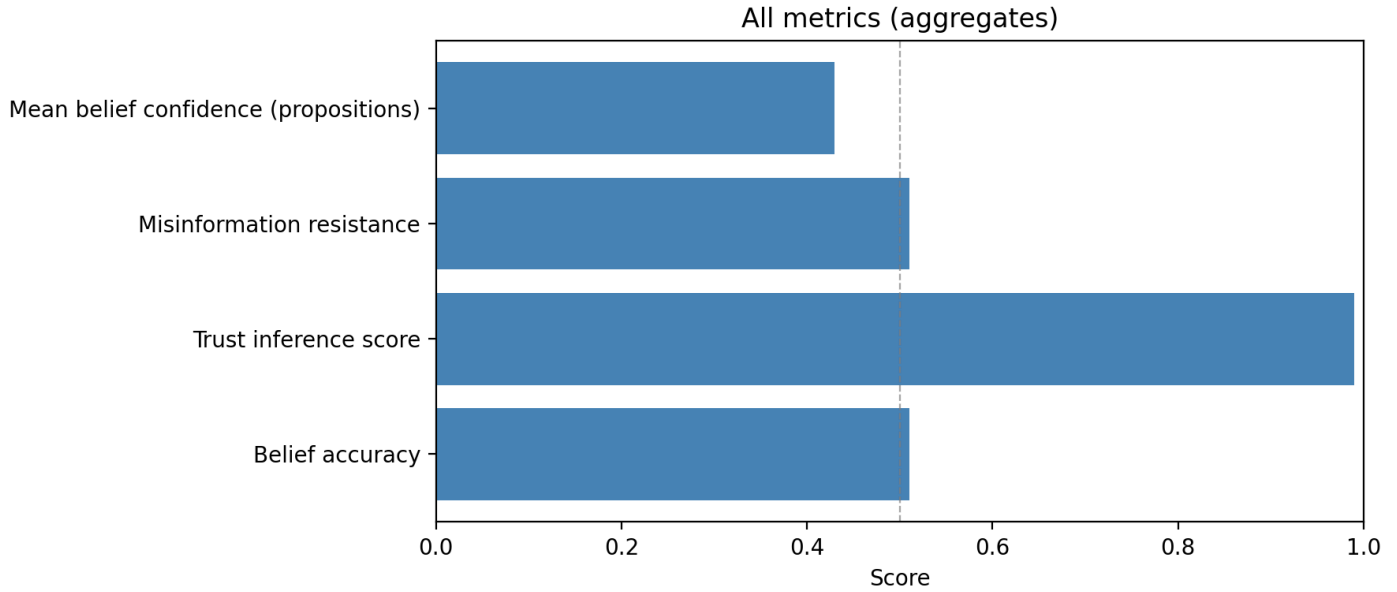


Figure 1.2

Figure 1.0 shows the source reliability plot. The three truthful agents all achieved perfect reliability, with every one of their statements counted as correct and none as incorrect. In contrast, the three deceptive agents produced the exact opposite pattern: all of their statements were incorrect, yielding a reliability score of 0.0. The noisy agents fell somewhere in the middle, with reliability values clustered around the midpoint, which is what would be expected from randomly generated claims. The strategic agents also remained around the midpoint in this run, indicating that their alternating behavior produced neither consistently high nor consistently low reliability overall. The bar heights show the raw counts of correct and incorrect statements, while the line plot summarizes those counts into a single reliability score for each source. Taken together, the figure shows that the reliability metric is behaving sensibly and is able to recover the intended behavioral differences among agents. The main implication of this result is that the system's source-tracking component is effectively identifying which agents can be trusted and which cannot. This is important because it means the experiment is not merely counting true and false statements, but also learning a meaningful profile of each informant over time. The truthful agents forming a clean upper bound and the deceptive agents forming a clean lower bound suggest that the design has strong discriminative power. Meanwhile, the noisy and strategic agents illustrate the harder middle case, where reliability is partial rather than absolute. That middle region is especially relevant for social reasoning, because real-world information sources are often neither fully honest nor fully deceptive.

Figure 1.1 is dedicated to the confidence evolution score and shows a steady increase in belief confidence across the propositions as more evidence is accumulated. Most propositions begin at the initial confidence level of 0.90, then rise quickly into the mid-0.90 range after repeated confirming statements are received. Over time, the curves converge near 0.99, which suggests that the system becomes increasingly certain about propositions that continue to receive support. The small dips and brief fluctuations early in the

sequence indicates that conflicting testimony does affect confidence, but it seems only temporarily. After enough corroborating evidence is observed, those fluctuations are mostly smoothed out and the beliefs stabilize at a higher confidence level. This pattern is consistent with an incremental reasoning process in which repeated agreement strengthens belief while isolated contradictions produce only short-term uncertainty. From this, I gathered that the system is not treating every new statement as equally decisive, but instead is gradually integrating evidence over time. That is a desirable property for reasoning in a socially uncertain setting, because it means the agent does not overreact to a single misleading source. At the same time, the graph also suggests that confidence tends to remain high once it has been established, even when some contradictory claims appear later. So in other words, the belief updates appear to favor accumulation and stabilization more so than sharp reversals. This seems to be useful for demonstrating robustness, although it also means the system may be somewhat conservative in reducing confidence after misinformation.

Figure 1.2 shows the aggregate metrics plot and provides a concise summary of overall system performance across the key evaluation metrics. The most notable result is the very high trust inference score, which is very close to 1.0, indicating that the system was highly effective at identifying which agents were reliable and which were not. This aligns with the earlier source reliability results and suggests that the system successfully learned to differentiate between truthful, deceptive, and partially reliable agents over time. In contrast, belief accuracy and misinformation resistance both fall just above the midpoint, around 0.5, indicating that while the system performs better than random guessing, it is still significantly affected by misleading and conflicting information. Since misinformation resistance is operationalized similarly to belief accuracy in this implementation, the close alignment between these two values is expected. The average belief confidence across propositions is slightly below 0.5, which is an interesting contrast to the confidence evolution plot. While individual propositions can reach very high confidence when supported consistently, the average confidence across all propositions is reduced by those that receive mixed or contradictory evidence. This highlights an important distinction between local and global behavior. Some beliefs become highly certain, but overall system certainty is moderated by ambiguity in the environment. The dashed reference line at 0.5 provides a useful baseline, showing that belief accuracy and misinformation resistance only marginally exceed chance-level performance, whereas trust inference significantly surpasses it. The key implication of these results of Figure 1.2 is that the system is much more successful at learning *who to trust* than at consistently determining *what is true*. This seems to suggest that while the source evaluation mechanism is strong, the integration of that information into final belief formation is still vulnerable to noise and deception. Basically, the system appears to correctly recognize unreliable agents, but that knowledge does not fully translate into improved belief accuracy under adversarial conditions. This gap points to a limitation in how source reliability is currently incorporated into reasoning, and it highlights an important direction for improvement: more effectively weighting or filtering incoming information based on learned trust.

Future Work and Conclusion

There are several natural directions to extend this project in a way that would make the experiment more realistic and more informative. One important next step I may consider is to move beyond the current static fact environment and allow the underlying world state to change over time, since real reasoning systems rarely operate in conditions where truth just stays fixed. That would make the task harder in a more meaningful way, because the system would need to distinguish between a source becoming unreliable and the world itself simply changing. Another useful extension would be to give NARS more structured feedback about source quality, or to compare the current setup against a version where source reliability is explicitly incorporated into belief updates, so that the effect of trust modeling can be tested more directly. Another idea I had in mind was to maybe expand the agent pool to include subtler forms of deception, such as agents that lie only about certain facts, agents that become reliable after a period of deception, or agents whose behavior depends on the behavior of other agents. It would also be valuable to run much larger experiments over longer time spans, because short runs can show the basic pattern of behavior but do not fully reveal whether trust learning remains stable over many interactions. In addition, the current metrics could be refined so that misinformation resistance is measured more specifically under adversarial conditions rather than as a near-duplicate of general belief accuracy. A more direct analysis of NARS's internal belief revision process would also strengthen the study, especially if future versions record the exact point at which confidence changes in response to particular statements. The last idea I am thinking about is to do a comparison between this NARS-based setup and another reasoning or trust-learning system that would help show whether the behavior observed here is unique to non-axiomatic reasoning or part of a broader pattern in uncertainty management altogether.

This project set out to test whether NARS could reason effectively in a socially uncertain environment where information is not only incomplete, but also unreliable or intentionally misleading. My hypothesis was that NARS would maintain relatively high belief accuracy while also learning to distinguish between trustworthy and untrustworthy sources over time. The results provide a mixed but informative answer to that expectation. On one hand, the system performed very well in identifying source reliability: truthful agents were consistently recognized as reliable, deceptive agents as unreliable, and noisy or strategic agents were placed appropriately in between. This strongly supports the second part of the hypothesis, showing that NARS, or at least the reasoning framework built around it, is capable of tracking patterns of consistency and adjusting trust accordingly. The high trust inference score reinforces the idea that repeated agreement strengthens credibility while repeated contradiction weakens it, which aligns closely with the intended behavior of non-axiomatic reasoning. On the other hand though, belief accuracy and misinformation resistance were more moderate than expected, remaining only slightly above baseline. This suggests that while the system can learn who to trust, it does not yet fully translate that knowledge into consistently accurate beliefs in the presence of adversarial input. In other words, misleading information still has a noticeable impact on the final belief state, even when unreliable sources are correctly identified. This partially challenges the first part of the hypothesis, which anticipated stronger robustness in maintaining correct beliefs under deception. However, the confidence evolution results do show that NARS updates beliefs incrementally and becomes more stable as evidence accumulates, which supports the underlying reasoning behind the hypothesis. The system does not overreact to individual false claims, and it

demonstrates a form of resilience through gradual evidence integration, even if that resilience is not yet strong enough to fully suppress misinformation.

Taken together, the results suggest that the core intuition behind the hypothesis is sound, but only partially realized in this implementation. NARS appears well-suited for adapting to the social context of information by learning source reliability, but less effective at fully leveraging that knowledge to maximize belief accuracy in adversarial settings. This distinction is important, because it highlights that robust reasoning in multi-agent environments requires not just identifying trustworthy sources, but actively incorporating that trust into the weighting of evidence. Ultimately, the experiment demonstrates that non-axiomatic reasoning provides a promising foundation for handling uncertainty that arises from both the environment and other agents, even if additional mechanisms are needed to fully achieve the level of robustness originally hypothesized.